From Area Models to Expanding Algebraic Expressions

PhET Area Model Multiplication · PhET

Year 10	**~30 minutes**	**Mathematics · Algebra & Functions**

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Simulation: https://phet.colorado.edu/en/simulations/area-model-multiplication
Curriculum links: AC9M10A01 · AC9M7A02

Learning intentions

- I can use an area model to break a multiplication into partial products and add them together.
- I can use an area model to expand a bracketed algebraic expression such as (x + 3)(x + 2).
- I can connect the area model to the distributive property.

Getting started

Open PhET Area Model Multiplication and start on the Multiply screen. Build a rectangle for a simple numeric multiplication first, for example 12 x 15, splitting the sides into tens and ones, before moving on to the Variables screen for algebraic expressions.

Tasks

1. Build a rectangle for 12 x 15 by splitting the sides into 10 + 2 and 10 + 5. List the four partial-product areas you get.

2. Add your four partial products from the previous task together. Does the total match 12 x 15 calculated the normal way?

Working:

3. Switch to the Variables screen. Build a rectangle with sides (x + 3) and (x + 2). What are the four partial-product areas, in terms of x?

4. Add the four partial products from the (x + 3)(x + 2) rectangle to write the expanded expression. Simplify by collecting like terms.

Working:

5. Build three different bracketed expressions of your choice and record the expanded, simplified result for each.

Expression	Side lengths used	Expanded (simplified) result

6. Explain, using the area model, why (x + 3)(x + 2) is not the same as x^2 + 6. Refer to the partial products you found.

7. Explain in your own words how the area model connects to the distributive property you may have used before with brackets.

Extension (optional)

Use the area model to expand (x + 3)^2. Explain why the two 'middle' partial products are equal, and what this means for the expanded form.

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